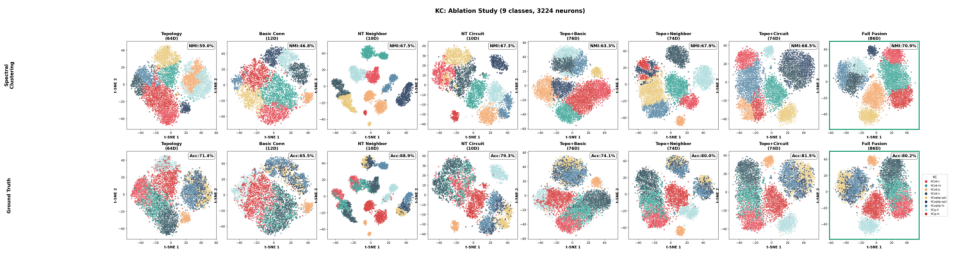
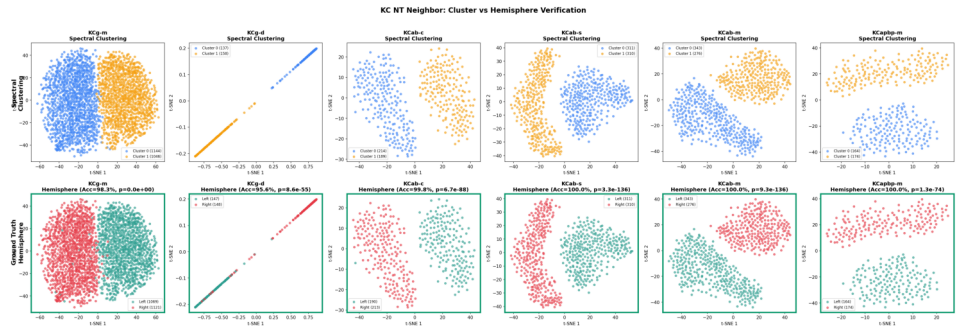
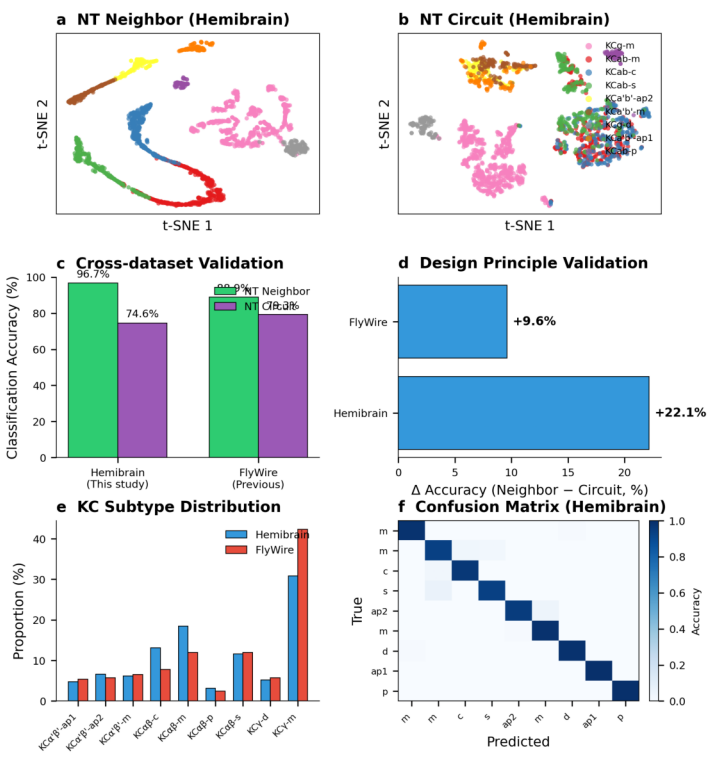
a FlyWire KC: Feature Ablation Study



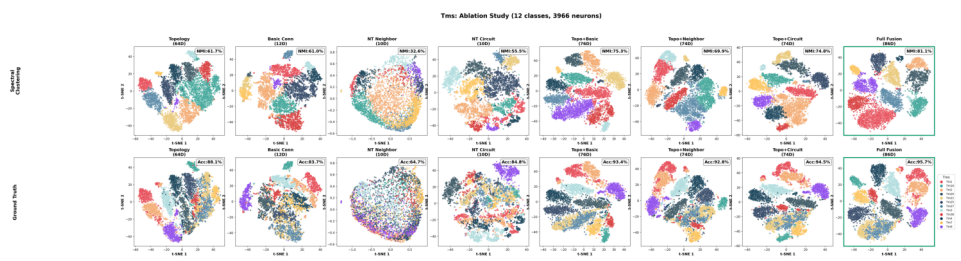
b FlyWire KC: Hemisphere Verification



c Hemibrain: Cross-Dataset Validation



d FlyWire Tm: Feature Ablation (Hierarchical Circuit)



e Design Principle Summary

Modulatory Circuits

- Mushroom Body (KC)
- Central Brain neurons

→ NT Neighbor (Local)

Best: 88.9% (KC)

Hierarchical Circuits

- Visual neurons (Tm)
- Optic Lobe pathway

→ NT Circuit (Global)

Best: 84.8% (Tm)

✓ Cross-Dataset Validated

FlyWire (bilateral): NT Neighbor +9.6% over NT Circuit

Hemibrain (unilateral): NT Neighbor +22.1% over NT Circuit

Design principle generalizes across independent datasets

f Classification Accuracy Summary

Category	Neurons	Topology	NT Neighbor	NT Circuit	Best Δ
KC (FlyWire)	5,177	71.4%	88.9%	79.3%	+9.6%*
KC (Hemibrain)	1,927	-	96.7%	74.6%	+22.1%*
Central	2,856	70.7%	79.6%	72.2%	+7.4%*
Tm (Visual)	3,015	88.1%	64.7%	84.8%	+20.1%†
Visual	17,893	96.9%	94.9%	95.5%	+1.4%†

* NT Neighbor > NT Circuit (Modulatory circuits prefer local context)

† NT Circuit > NT Neighbor (Hierarchical circuits prefer global context)